

Week 5 Initial Analysis Report

Track: Data Analytics Author: Brian Kariuki Program: AnalystLab Africa

1. Executive Summary & Data Engineering Summary

In Week 5, the raw 5,000-record HealthConnect dataset underwent an extensive data cleaning and quality audit pipeline. Continuous missing variables (clinic distance and wait times) were imputed using subgroup medians to preserve distribution shape. Categorical missingness was explicitly assigned, and date columns were cast to standard ISO datetime objects. All transformations were written out to a clean secondary file: `HealthConnect_Appointment_Data_Cleaned.csv`.

2. Calculated Baseline Key Performance Indicators

KPI Metric	Value	Operational Context
Overall No-Show Rate	48.46%	2,423 missed visits out of 5,000 total appointments
Reminder Attendance Uplift	+4.95%	Sent: 47.63% vs Not Sent: 42.68%
Avg Booking Lead Time Gap	24.52 vs 34.53 Days	Attended vs No-Show appointment booking gaps
High Wait (>30m) No-Show Rate	47.67%	Historical drop-off rate when clinic wait exceeds 30m

3. Data Preparation & Cleaning Audit Log

Field Name	Data Type	Quality Issue	Imputation Strategy
distance_to_clinic_km	Float64	90 missing values	Clinic location group median imputation
waiting_time_minutes	Float64	60 missing values	Service type group median imputation
reminder_channel	Categorical	1,366 missing values	Encoded as 'No Reminder Sent'
appointment_date	Datetime64	String formatting	Standardized ISO datetime object